

Web-Based Negotiation Tool for Conflict Resolution in Higher Airspace Operations

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In higher airspace operations, vehicles such as high-altitude, long-endurance balloons and fixed-wing gliders have diverse characteristics and limited maneuverability. To handle vehicles with these unique characteristics, a new strategic negotiation-based conflict resolution method with a real-time web-based negotiation tool has been proposed to avoid potential conflict between vehicles in Upper Class E airspace. To facilitate negotiation, this tool incorporates a bilateral negotiation model with an algorithm to assess conflict risks and generate new flight trajectories with a given flight path deviation. This tool allows users to make decisions during negotiation, both manually and automatically, while enforcing the necessary constraints for successful negotiation. An experimental study is carried out to evaluate the impact of varying negotiation strategies, vehicle types, crossing angles, and operator response times during negotiation on metrics such as total completion time, number of negotiation rounds, and extra flight distance to avoid conflict. The results suggest that operators benefit from using negotiated flight paths with quick response times across various crossing angles and vehicle types. Compared to the conventional method, where a single operator assumes full responsibility, the findings suggest that negotiation-based strategic deconfliction reduces the total extra flight distance by an average of 25%.

Nomenclature

c_i	=	cost associated with operator i
$d(f_i)$	=	flight path distance of a trajectory for operator i
f_i	=	trajectory associated with the i th operator, where $i = 1, 2$
$f_{i,B}$	=	baseline trajectory for operator i
$f_{i,P}$	=	proposal trajectory for operator i
$f_{i,R}$	=	response trajectory for operator i
$f_{i,0}$	=	original trajectory for operator i
β	=	negotiation strategy value for operator i
δ_i	=	normalized flight distance for operator i

I. Introduction

UPPER Class E airspace is defined as over 60,000 feet (or flight level/FL600) in the National Airspace System (NAS). There is a projected increase in demand for operations that must effectively maneuver in this airspace. Ensuring safety for aircraft in Upper Class E is a unique challenge because of the vast diversity of aerial vehicle

types operating in the near future. Such aircraft include supersonic transport vehicles, high-altitude, long-endurance (HALE) fixed-wing vehicles, HALE balloons, high-speed unmanned fixed-wing aircraft, HALE airships, and other vehicles designed for space launches. These vehicles operate in or pass through Upper Class E airspace [1]. Each vehicle has different levels of maneuverability, speed, weight, size, flight duration, and control, creating unique challenges for these vehicles to engage in conflict detection and de-escalation.

In anticipation of increasing demand for operations in Upper Class E airspace, NASA and the FAA are exploring a new concept called Upper Class E Traffic Management (ETM). In the current proposal plan by the FAA, Version 1.0 ETM Concept of Operations [2], operations in Upper Class E airspace require cooperative measures with operators managing and sharing their operational intents with neighboring operators to avoid conflict. There is a consensus in the community that negotiation for strategic deconfliction is the right direction, but there are no specific guidelines for what this might look like. There is also an active research question regarding the equity and trust of the cooperative measures within ETM.

In response to several open questions from industry partners and the FAA, the authors in this paper have introduced a novel negotiation model that is tailored to meet the needs of ETM vehicles to avoid potential conflict between vehicles well in advance of the conflict point and with ample time for bilateral negotiation [3]. This model assumes that operators are willing to cooperate and share their operational intents, but the details of an operator's mission, such as true cost, can be kept private. Using principles from game theory, this model algorithmically generates equitable, conflict-free flight trajectories within minutes. The solution set of conflict-free trajectories found through negotiation is mutually agreed on and equitable since all operators share the responsibility and absorb partial costs. An initial evaluation of this model has suggested that outcomes are based mostly on cost, strategy, and response times, and they are not affected by vehicle maneuverability or who proposes first.

The negotiation model has been refined and developed further since then. This paper describes the original negotiation model proposed in [3], as well as the improvements made to both the algorithm and the tool. One of the improvements is an update to the negotiation model, which changes the cost definition to utilize extra

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flight distance, instead of deviation, as a more intuitive and user-friendly approach to distributing maneuver responsibility. Another improvement is the development of a web-based user interface (UI) as a research tool to better understand the feasibility and equity of managing cooperative measures through an automated proposal and counterproposal approach. The UI has been developed to facilitate real-time interactions with the model and visualize the negotiation processes step by step. The tool consists of a front-end UI in web-based client software and the conflict-detection and trajectory generation algorithms in the backend, server-side software. The front-end UI allows human operators to interact and negotiate with other operators at each negotiation step, set the negotiation parameters, and visualize the outcome. This implementation allows the negotiation process to happen more realistically—with breaks between proposals and counterproposals that mirror human decision-making and potential delays caused by distributed web-based tool interactions similar to the ETM concept's possible implementations. Automated testing has been developed to conduct a sensitivity analysis and enable real-time simulations with this web-based negotiation tool. Various factors affecting the negotiation process are investigated through different scenarios, such as varying crossing angles, aircraft types, response times, and negotiation strategies.

The UI may also provide a template for a future tool that can support a human-centric negotiation process during the early implementations of ETM operations. Human-centric negotiation may be prevalent in the initial ETM, operating at a low traffic density, but as the concept matures and traffic increases, the negotiation is expected to move away from human-centric to automated processes. A negotiation model with an interactive UI may initially provide model transparency and insights for operators on the type of negotiation being executed and whether it is appropriate. Those insights may be used to gradually and iteratively refine the negotiation process until there is sufficient trust in the model to transfer the negotiation process completely to automation.

Further refinements include a more sophisticated methodology for performing experiments through an automated approach. The previous model performed experiments under a fast-time analysis simulated in a singular MATLAB environment responsible for the input operations of two different conflicting operators. Our current model utilizes a dual web-browser environment, with two separate instances of the model running and communicating with each other through a client-server architecture setup, rather than in a single offline client. With the dual web-browser environment, an automation system is developed that physically manipulates the two separate web browsers, where usually a human is required to manually interact with the UI, enter costs, adjust sliders, and click buttons. The automation system automatically handles the web browser's interactive elements while considering different input parameters such as an operator's flight plan, response time, and negotiation strategy. Lastly, several bug fixes and optimizations have been applied to the algorithmic implementations of the trajectory generation and negotiation model systems, which increases the speed and reliability of the model.

The remainder of this paper is organized as follows: Section III reviews the existing literature on the negotiation models, conflict detection algorithms, and trajectory generation algorithms. Section IV details the negotiation model algorithm implemented in this paper, Sec. V describes the tool UI development, and Sec. VI walks through the process for automating human interactions required to conduct a sensitivity analysis. Then, in Sec. VII, the experimental details, such as the input factors and output measures, are covered. Results and the sensitivity analysis for key parameters that impact the effectiveness of the negotiation model are analyzed in Sec. VIII. Key observations from the sensitivity analysis are described in Sec. IX. Concluding remarks and the direction of future work are found in Sec. X.

II. Review of Existing Work

This section reviews the literature on two-party negotiation models for air traffic management and conflict risk assessment and reduction algorithms.

A. Negotiation Models for Air Traffic Management

Negotiation has been widely used as a way for systems or agents to resolve conflicts and come to agreements. Rubinstein's foundational bargaining model formalized two-party negotiation over time. His work showed how rational agents with time preferences could reach equilibrium agreements under deadline pressure [4]. Building on this, Faratin et al. proposed a framework for autonomous agents to make negotiation decisions based on utility functions, concession strategies, and similarity heuristics. It provided a flexible basis for agent-based negotiation in dynamic environments [5]. Parsons et al. introduced argumentation-based negotiation, where agents do not just accept or reject offers but also exchange explanations and constraints to refine proposals and reach mutually acceptable solutions iteratively [6]. These ideas have been applied in many areas, including e-commerce and transportation.

These foundational ideas have also shaped how negotiation is applied in more specialized domains, such as air traffic management, where agents must coordinate in real time under safety-critical conditions. One of the main challenges in the proposed ETM concept of operations is ensuring a minimum separation between aircraft without the guidance of a third-party agent in a free-flight setting. Several negotiation models have been developed for applications in the air traffic management domain to maintain minimum separation in a free-flight setting. There are two categories of negotiation models: non-cooperative and cooperative models. Non-cooperative models rely on a non-negotiated outcome of one operator bearing the full burden to maneuver while the opposing operator does not contribute to deconfliction [7]. Building on this, Bilimoria introduced a geometric optimization approach that solves pairwise conflicts via convex programming without explicit bargaining [8]. Cooperative models take on a negotiating approach, where operators propose and counterpropose with conflict-prone opponents until a solution is developed [9,10].

This paper uses a negotiation model based on a cooperative framework, drawing inspiration from prior research. The work from Menon et al. [11] uses a static negotiation model approach, where operators are assumed to generate a full list of candidate flight plans associated with scores and share them to form a utility matrix, from which Nash Equilibrium [12] can be identified and a solution can be selected from it. Pritchett [13] developed a negotiation model based on the sequential bargaining model for tactical conflict resolution among air traffic. Revised trajectories consisting of heading changes (turn left/right), altitude changes (climb/descend), and speed changes (acceleration/deceleration) are required. Time constraints are used to ensure that the negotiation process can finish within a given number of steps. Rong et al. [14] proposed a model where each aircraft acts as an agent. The aircraft negotiate directly with each other in pairs. If a proposal is rejected, the other aircraft explains which rule it breaks. This helps both sides find a better solution without sharing private details. The system can handle many conflicts at the same time, and an air traffic controller only steps in if the aircraft cannot agree in time.

Our previous work [3] proposes a sequential bargaining negotiation model for strategic planning in high-altitude operations built upon sequential bargaining negotiation. From measuring the total available negotiation time, a time-dependent utility is developed to incentivize reaching an agreement by the end of the negotiation deadline. This concept is implemented in this paper as required cost, which helps ensure operators converge closer toward a solution. Due to vehicles having different maneuverability and operator preferences, several input parameters must be considered for negotiation. To reduce input complexity, the flight trajectories are proposed based on normalized deviation, producing a left and right heading flight trajectory instead of multiple dimensions as in [13]. The definition of the normalized cost ensures that operators can express their actual needs via different preference setups without exposing their private business costs.

B. Conflict Risk Assessment and Risk Reduction Algorithm

Due to the predictive uncertainty associated with the high-altitude vehicle positions, the strategy for assessing the conflict risk between

two operators involves measuring the probabilistic chance of conflict. Previous studies have described different methodologies and algorithms to detect conflict. Two papers [15,16] in particular describe two analytical methods for assessing conflict probability between two parties. Paielli's work [15] calculates the likelihood of conflict based on the amount of overlap between the position errors of both vehicles, using a constant relative velocity. The main drawback is that the approximation may become inaccurate in the presence of a variable speed or during turn segments. Hwang's approach [16] refines Paielli's approach by evaluating the conflict as a set of instantaneous conflict probabilities, approximating the relative velocity elliptical conflict zone as a rectangle, and measuring the overlap at a given time. This reduces the inaccuracies that occur with variable vehicle speeds and turn segments. The algorithm used in our earlier and current research [17] leverages a hybrid model that combines elements of both Paielli and Hwang's methods to enhance the accuracy of conflict risk assessment for turning segments, while also improving the accuracy of risk detection for missions with longer look-ahead times. The contribution made with the hybrid method adds a greater fidelity of accuracy for turn segments by switching from a normal distribution for the probabilistic position of a vehicle to a more sophisticated method that creates a bending error ellipse at turn segments rather than a straight error ellipse. This algorithm allows the negotiation model to input the flight trajectory plans and vehicle parameters from two operators and output a probabilistic likelihood of conflict as a percentage.

In addition, the negotiation model requires a deconfliction algorithm that can work in conjunction with the conflict detection algorithm to reduce conflict risk, in this case, deviating from the original trajectory. The heading change algorithm is implemented alongside a bisection algorithm, developed in [8,18]. This combination generates a trajectory with extra flight distance onto an operator's original flight trajectory but with reduced conflict risk. Further research can explore extending these methods to incorporate other trajectory maneuvers.

III. Implementation of the Trajectory Generation, Conflict Detection, and Assessment Algorithms

There are various approaches to resolving conflicts via trajectory negotiations. However, this paper proposes a trajectory negotiation model comprising three categories of trajectories: baseline trajectories, proposal trajectories, and response trajectories. Baseline trajectories, $f_{i,B}$, represent an operator's maneuver responsibility to fully resolve conflict for one operator with no contribution from the other. A proposal trajectory, $f_{i,P}$, assumes a user-defined partial maneuver, expressed in the form of c_i and normalized into δ_i , which is complemented by a complete solution from the opposing operator's response trajectory, $f_{j,R}$, which resolves the remaining responsibility needed to fully deconflict.

A significant feature from our previous negotiation tool comes from the capability of producing a user-defined $f_{i,P}$ in the form of a cost c_i . The implementation of the trajectory generation from our previous work system calculates c_i by measuring the closest points of approach (CPA) between a newly generated flight trajectory plan and the opposing operator's original flight trajectory. One of the requirements of the negotiation model is that c_i must monotonically increase the CPA distance until it exceeds the minimum separation distance by the end of the negotiation window, which ensures that the negotiation process moves closer to an agreement at every round.

The physical meaning of c_i has been modified in the current implementation. During earlier testing of c_i , it has been found that deviation based on CPA did not always meet the requirement of a monotonically increasing cost. As a result, this brings potential issues toward a potential convergence if operators are not working closer to a solution at each time step. To remedy this issue, this paper utilizes flight distance as an additional layer on top of the deviation algorithm within the trajectory generation system to provide a more intuitive and reliable behavior toward achieving a monotonically increasing c_i . Previous work uses a CPA-based metric called

deviation in tandem with c_i as a measurement of maneuver responsibility. This paper introduces a new metric that measures extra flight distance called the normalized extra flight distance δ_i . It still uses CPA as guidance to ensure that aircraft maintain a minimum separation of 5 nmi, but rather than relying on a metric that measures CPA distance, it measures extra flight distance instead. The updated terminology in this paper replaces deviation with δ_i and is defined in Eq. (1). The flight distance of a trajectory, denoted by $d(f_i)$, is used to calculate the extra flight distance ratio between an operator's updated maneuver and their full maneuver. The numerator represents the extra flight distance of their partial maneuver. This is calculated by the difference between the partial maneuver, either $f_{i,P}$ or $f_{i,R}$, and their nonmaneuver plan $f_{i,0}$. The denominator represents the extra distance calculated by the difference in flight distance between the full responsibility maneuver $f_{i,B}$ and the nonmaneuver $f_{i,0}$. Therefore, the definition of δ_i describes the amount of responsibility an operator inherits to avoid conflict in terms of flight distance.

$$\delta_i = \frac{d(f_{i,P} \vee f_{i,R}) - d(f_{i,0})}{d(f_{i,B}) - d(f_{i,0})} \quad (1)$$

Another inherited feature the trajectory generation system needs to consider is an operator's negotiation strategy. This is represented by the β value, which represents the maneuver intent based upon a function of time. Previous research utilizes different β values to represent different negotiation strategies; for example, the Boulware strategy, which is represented as $\beta \gg 1$, employs a negotiation strategy in which an operator is hesitant to maneuver early in the negotiation process from $f_{i,0}$ until the remaining negotiation time is low. The conceder strategy, represented as $\beta \ll 1$, represents an operator much more willing to maneuver from $f_{i,0}$ at the start of a negotiation. When $\beta = 1$, the relationship between c_i and δ_i is linear and one-to-one throughout the negotiation process. Previous research efforts utilize β as a time-dependent relationship between c_i and deviation. In this paper, since a CPA-based cost calculation is replaced by one that uses extra flight distance, the equation that represents the relationship between c_i , δ_i , and β is updated, as shown in Eq. (2).

$$c_i = \delta_i^{1/\beta} \quad (2)$$

Given the trajectories from the two conflicting vehicles, the conflict risk assessment method assesses and observes the highest conflict probability in the simulated operational intent and when it occurs. The input parameters that the conflict risk assessment algorithm considers are the vehicle details of both operators, including vehicle speed, vehicle model, turn rate, and max bank angle. Both operators' vehicle details and flight trajectories are run through a simulated operational intent that checks the distance of both aircraft throughout the simulation. If the distance between the two aircraft is less than five nautical miles during the simulation, then the severity for each instance of conflict and its corresponding time are recorded. From the output of the simulated operational intent, the four metrics observed are the conflict probability when conflict is first introduced, the time when conflict is introduced, the peak conflict probability, and the time when conflict is at a peak. The first two metrics are used to determine the total available negotiation time, and the last two are used to aid in creating conflict-free trajectories. If the goal is to produce a conflict-free trajectory for an operator, then that is achieved by rerouting an operator's flight trajectory until it produces a conflict probability with the opposing operator of 0%. If the conflict risk assessment algorithm is rerun with the modified trajectories and determines that the peak conflict probability is 0%, then the given trajectories are conflict-free.

To describe how the trajectory generation and conflict risk assessment algorithm are integrated into the negotiation tool, suppose an operator's flight trajectory is generated as a proposal flight trajectory $f_{i,P}$. Since $f_{i,P}$ is generated based on a user-defined proposal cost c_i , there is no guarantee that the $f_{i,P}$ is conflict-free with the opposing

operator's original flight trajectory $f_{j,0}$. To guarantee a conflict-free resolution, the trajectory generation system modifies the opposing operator's $f_{j,0}$ until it outputs a conflict-free response trajectory $f_{j,R}$. This involves an iterative process with the trajectory generation system to find an $f_{j,R}$ so that $f_{i,P}$ and $f_{j,R}$ are conflict-free with each other. The negotiation model calculates a response flight trajectory $f_{j,R}$, which takes the opposing operator's $f_{j,0}$ and maneuvers in both a left and right turn heading change until $f_{j,R}$ becomes conflict-free with $f_{i,P}$. For each potential $f_{j,R}$ solution resulting from the trajectory generation algorithm, the conflict risk assessment algorithm calculates the peak probability of conflict. If the peak probability of conflict is greater than 0%, then it signals the trajectory generation algorithm to increase the heading turn angle needed to generate $f_{j,R}$ and reassess. If the peak probability is 0%, then there is no need to rerun the trajectory generation algorithm since the $f_{j,R}$ solution is conflict-free with $f_{i,P}$. The iterative process ends when both $f_{i,P}$ and $f_{j,R}$ are conflict-free, and only in that condition can a response cost be calculated.

IV. Design and Development of Web-Based Negotiation Tool

In our previous research, Xue, Ishihara, and Lee have proposed, developed, and evaluated a two-way negotiation model for ETM operations [3]. Although the results were positive and promising, the method could only be tested in a fast-time analysis, and it did not allow a way to interact with the model in real-time or to participate in and visualize each negotiation step [3,14]. However, there is no easy method for generating the interactivity of two operators inherent in the concept of negotiation. In this research, a web-based tool is developed to address this problem by building a user interface (UI) that allows two operators to connect, communicate, and negotiate with each other in real-time.

The tool allows human operators of the impacted vehicle(s) in conflict to access the negotiation model and its parameters interactively in real time. It has a distributed architecture, with the front-end UI in web-based client software and the negotiation algorithms and models in the backend, server-side software. The front-end UI allows human operators to interact and negotiate with other operators at each negotiation step, set the negotiation parameters, and visualize the outcome. This implementation allows the negotiation to happen more realistically—with breaks between proposals and counterproposals that mirror human decision-making and potential delays caused by distributed web-based tool interactions similar to the ETM concept's potential implementations. Our negotiation tool comprises several functions to facilitate negotiation rules, communication between operators, conflict detection, and trajectory generation. Additional capabilities are built to run the tool in an automated batch mode to test a large set of simulation runs using the same tool built for human-in-the-loop operations.

A. Architecture Design

The client interacts with the program via a web UI, as shown in Fig. 1. Users can control the program through a two-way Web-Socket channel [19] with a Python-based negotiation server, which handles the negotiation model's data generation, conflict detection, data storage, and trajectory generation. The REST API [20] facilitates communication between the client and server. The program's logic is organized into modules, as shown in Fig. 1. Function calls represent the input and output flow of data between the various algorithms that handle conflict detection, data storage, trajectory generation, and the web UI, with the negotiation server. The Web-Socket events and API calls represent the bilateral communication between the negotiation server and the web UI.

B. Design of User Interface

To briefly walk through the UI, once the operators initiate the tool, they are greeted with an initial setup, where each operator can input their username, conflict group, flight trajectory, and negotiation strategy, as shown in Fig. 2. The negotiation strategy is

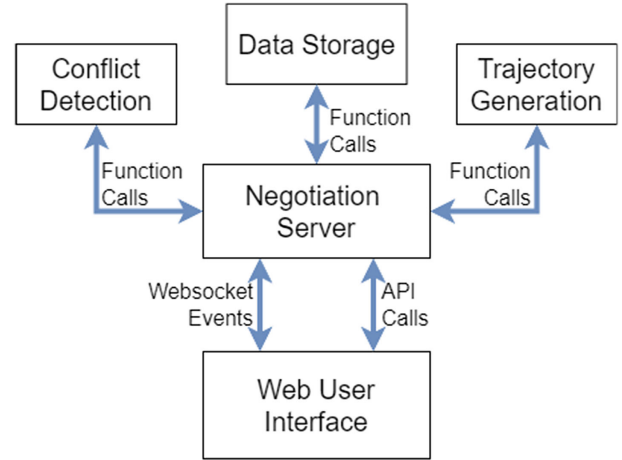


Fig. 1 Simple architecture design diagram for the negotiation tool.

Fig. 2 Setting up a conflict scenario for testing.

expressed through the beta strategy slider, whose value impacts the cost versus normalized flight distance curve, as shown at the bottom of the same figure. Larger beta strategy values cause the line to curve upward, as shown in Fig. 2, which allows users to express their intent to give in more maneuver responsibility later on in the negotiation, and lower beta values cause the line to curve downward, allowing users to express an intent to give in a large maneuver toward the beginning of negotiation. The conflict group is used for internal testing, which can be ignored. Once both operators enter their initial inputs and click “Submit Operation,” the UI sends the information to the negotiation server to check for conflict. If conflict is detected, a popup alert indicates to the operators that conflict exists, and negotiation begins. During negotiation, relevant conflict information would appear on the right side of their screen, as shown in Fig. 3, which displays the amount of time to negotiate, the required cost (the time-dependent utility to ensure operators work closer toward a solution at each time step), the conflict peak probability, and a visual map of both operators's flight trajectories.

The first proposer sees a proposal UI as shown in Fig. 4 to enter their c_i . The program uses their negotiation strategy β value to

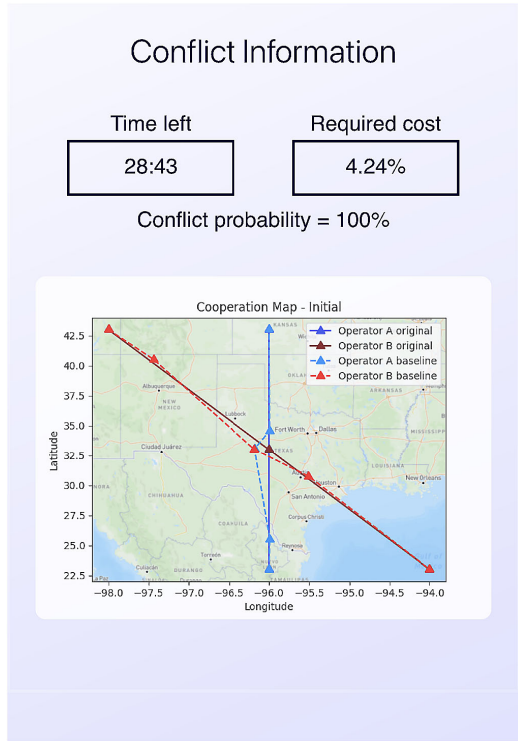


Fig. 3 The relevant conflict information displayed during negotiation.

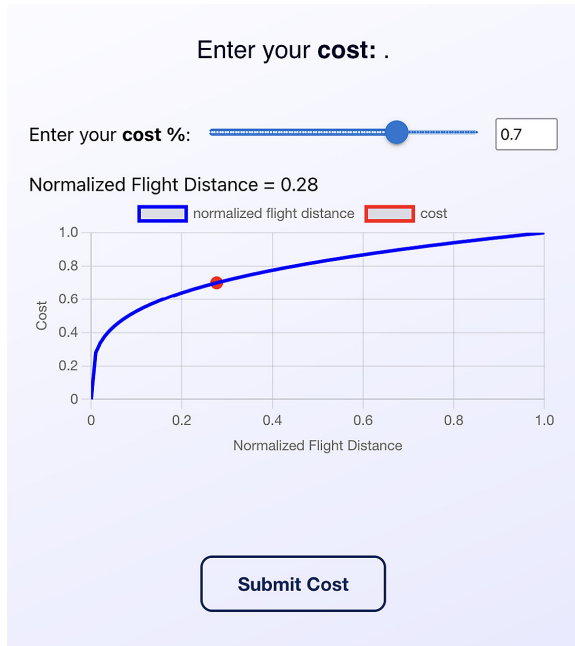


Fig. 4 The UI to enter the cost of a proposal trajectory.

calculate δ_i . After submitting the cost in Fig. 4, the trajectory generation algorithm computes a proposal trajectory $f_{i,P}$ based on the desired normalized flight distance δ_i . The proposer can review their $f_{i,P}$, as shown in Fig. 5, before sending the trajectory to the other operator.

After the proposing operator sends their trajectory $f_{i,P}$, the responding operator receives an updated overview map with $f_{i,P}$, shown in Fig. 5, and generates a response trajectory $f_{j,R}$ by clicking “Generate Response.” This triggers the trajectory generation algorithm, producing $f_{j,R}$ and c_j . Following that initiates the final step of the UI, and the following trajectories are shown on the map as shown in Fig. 6: $f_{i,P}$, $f_{j,R}$, $f_{i,0}$, and $f_{j,0}$, alongside

Trajectory with cost = 70% and normalized flight distance = 27.69%

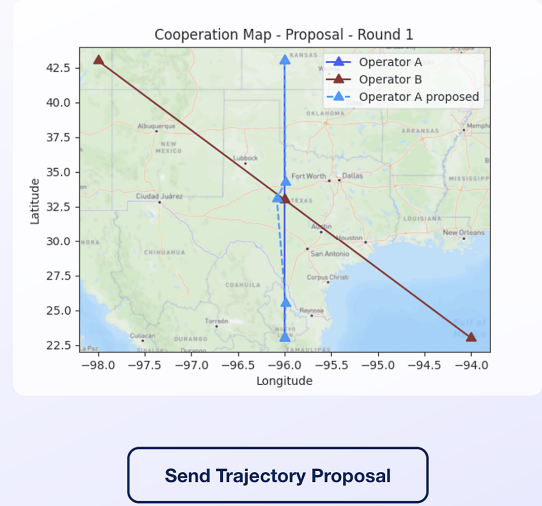


Fig. 5 The proposal trajectory UI.

Generated response trajectory below, with response cost = 29.6%



Fig. 6 The response trajectory UI.

the responder’s c_j . At this step, the responding operator can decide to “Accept” or “Decline” based on their preference as long as the response cost is greater than or equal to the required cost. If the response cost is less than the required cost, the responding operator can only choose the “Accept” button. The restriction of the decline button in this case is implemented as a convergence mechanism. If the response cost is less than the required cost, then it is unlikely that the operator would get a better deal from further negotiating. Therefore, the operator must accept the plan. If the response cost is greater than the required cost and “Decline” is chosen, the roles swap, starting a new negotiation round with the new proposer entering a cost. This cycle continues until a responding operator clicks “Accept,” resulting in mutually agreed upon conflict-free flight

trajectories, or a responding operator undergoes the convergence condition that their response cost is less than the required cost and is forced to accept.

V. Automating Human Interactions

The web-based negotiation tool enables human operators to interact with the negotiation algorithm via a web-based user interface that visualizes flight trajectories and allows for real-time adjustments. The tool architecture poses challenges for conducting batch runs necessary for exploring negotiation strategies and sensitivity analyses. Each experimental run can take between 30 min and 2 hours, with hundreds of runs needed for a comprehensive study, leading to extensive manual work.

To address this, an automation system is developed to create a controlled environment for sequential batch processing with pre-configured input parameters. It accounts for natural delays in proposal and response interactions, which impact results and were not implemented in prior fast-time simulation studies. Additionally, it streamlines this by adjusting under-defined input parameters such as response time throughout the negotiation process.

The automation system is developed using the Selenium Python package [21], enabling the simulation of web-based interactions for sensitivity analyses. It can simulate user actions within the negotiation tool through two types of interactions: 1) physical interactions, such as inputting information, selecting items from drop-down menus, and clicking buttons, and 2) decision-making actions, which involve evaluating contextual key data, such as time left in negotiation and required cost to process an action. Each simulated user operates independently, with their decisions prompting corresponding evaluations and actions from the other operator, creating a simulated dynamic negotiation environment.

A full negotiation cycle can be automated using a dual-browser setup generated with the Selenium Python package. This process involves two simulated users, with the left browser designated as the first proposer. Each automation batch contains unique input variables affecting the negotiation outcomes. The proposer calculates a proposal cost by adding 1% to a real-time required cost retrieved from the web browser. After entering this cost, the tool sends the generated proposal plan and switches the focus to the right browser, designated as the responder. The responder evaluates the proposal information based on the calculated response cost and the required cost. If the response cost exceeds the required cost, the responder declines the proposal, switching roles for a new round. If it is lower, the proposal is accepted, ending the negotiation. The automation system follows this structured mechanism, streamlining the sensitivity analysis process, allowing for repeatability, and eliminating potential human errors within each experiment.

VI. Applications of the Tool and Design of Experiment

This section will discuss the negotiation tool's applications and introduce the design of experiments to empirically demonstrate its capabilities. We will also outline the information we collected from each experiment, conduct a sensitivity analysis, and demonstrate the success of the developed tool in the next section.

A. Applications

This tool can be used in various scenarios specific to ETM conflict resolution. It is assumed that the operators using this tool are willing to share their flight trajectories to cooperate and that sufficient time is given to negotiate. The initial purpose of this tool was for strategic rather than last-minute, tactical deconfliction, to provide an efficient and equitable way to solve conflict between two operators with sufficient time for negotiation. The tool can simulate and evaluate a wide variety of inputs and outputs, thereby potentially uncovering interesting relationships between various vehicles, conflict geometry, and other environmental factors, and the resultant negotiated solution that could lead to complicated or delayed negotiation under specific input conditions. The results of simulation and

web-server interactions are generated in real-time using a 12-core MacBook with an M2 Pro chip and 16 GB of RAM. Each scenario is conducted with a time until conflict of 90 min, with 30 min subtracted as a negotiation fallback measure, resulting in a minimum of 60 min to negotiate between operators across all scenarios. The time to negotiate ranges from 60 min to 120 min, depending on the factorial design and input variables. Running a batch simulation through a controlled automation setting allows us to set up multiple run conditions with different input factors. Detailed information about these input factors, including how they relate to ETM scenarios, is explained next.

B. Factorial Design and Input Variables

We choose to test each operator's input variables to observe the effect of their interactions, ensuring that the negotiation is successful in many different scenarios. Table 1 summarizes the factors and levels chosen for experimentation. The following subsections explain each factor we are testing in more depth.

1. Crossing Angles

The crossing angles between two aircraft trajectories can categorize different conflict scenarios. Based on the range the angle falls into, the type of conflict is categorized. If the angle between the two aircraft is acute, between 0° and 45° , the case is classified as a shallow crossing conflict. The conflict is categorized as a crossing if the angle between the two trajectories is between 45° and 135° . The case is classified as a head-on conflict if the angle between the two trajectories is obtuse, between 135° and 180° . Refer to Table 2 for a quick summary.

2. Aircraft Type

The vehicles used in these experiments are meant to be as close as possible to those in the real world. To test how the tool can handle aircraft of various speeds and maneuverability in reality, the aircraft and trajectories are modeled after real vehicles that can operate in Upper Class E airspace. The two unique vehicles chosen for testing are the Global Hawk RQ-4 [22] and the Sun-glider solar-powered high-altitude pseudo-satellite (HAPS) unmanned aerial vehicle (UAV) [23]. The maximum bank angle used for the Global Hawk RQ-4 is reported to be 14.8% based on performance analysis and flight testing [24]. The speed for the Global Hawk reported by the U.S. Air Force [22] is 310 knots. The Sun-glider HAPS has been found to have an average cruise speed of 68 mph [23]. The summary of these parameters for the vehicles can be seen in Table 3.

Table 1 Factors, levels, and abbreviations included in testing

Factors	Abbr.	Levels
Crossing angles	C	Crossing
	SC	Shallow crossing
	H	Head on
Aircraft type	G	Global Hawk RQ-4
	S	Sun-glider HAPS UAV
Response time	I	Instant
	L	Long
Negotiation strategy	B	Boulware
	L	Linear
	C	Conceder

Table 2 Summary of conflict scenarios categorized by crossing angles

Category	Angle range
Shallow crossing	0° – 45°
Crossing	45° – 135°
Head on	135° – 180°

Table 3 Summary of studied vehicle parameters

Aircraft	Max bank angle	Avg. cruise speed	Max turn rate
Global hawk	14.8%	357 mph	0.93°/s
Sunglider	5%	68 mph	1.62°/s

3. Response Time

Response time pertains to the time it takes an operator to complete their response for any specific step in negotiation. In previous studies [3], it was found that increased response times can disadvantage the slower operator in negotiation. We are interested in extending the previous studies by adding in more input variables and testing whether or not that claim holds true. The summary for the response time parameters can be found in Table 4.

4. Negotiation Strategy

Similar to response times, we are also interested in extending the previous studies' hypothesis to see whether different negotiation strategies have a meaningful impact on the final output measures. The negotiation strategy β values for the Boulware ($\beta \gg 1$), linear ($\beta = 1$), and conceder ($\beta \ll 1$) strategies were chosen to be distinguishably different from each other, which might result in unique normalized flight distance behaviors among the three strategies. The summary of the negotiation strategies used within our experiments can be found in Table 5.

C. Output Measures

The following section briefly describes the output metrics measured at the end of a negotiation. These output metrics will be used to conduct a sensitivity analysis with the input variables to determine how different input combinations can affect the outcome of a negotiation.

1. Number of Rounds

The number of rounds refers to the number of full negotiation cycles, each of which consists of the flight trajectories from a proposing operator and a responding operator.

2. Total Time

Total time refers to the time it takes for a negotiation to finish from the start to the end of a negotiation.

3. Ratio of the Extra Flight Distance (Sum Ratio)

The ratio of the extra flight distance, or sum ratio, is calculated as the summation of the extra flight distances for *both* operators divided by the baseline extra flight distance for operator *A* or *B*. The baseline extra distance is the extra distance that would have been incurred if operators *A* or *B* had taken on 100% of the maneuver responsibility for resolving conflict.

Table 4 Summary of response times by duration

Category	Response time, s
Instant	1
Long	600

Table 5 Summary of negotiation strategies by category

Category	Strategy type
Boulware	$\beta = 4$
Linear	$\beta = 1$
Conceder	$\beta = \frac{1}{4}$

VII. Experiment Results

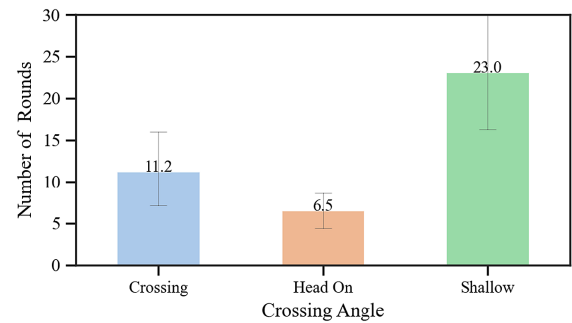
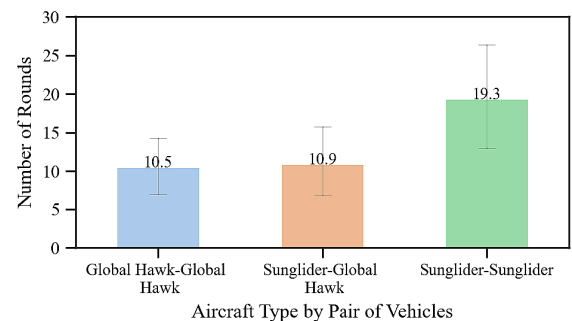
We are interested in investigating the relationship between each input parameter and the outcome of the experiments. A sensitivity analysis is conducted using our conflict detection and avoidance algorithms (CD&A), with a variation in the input variables, which include the crossing angle, aircraft types, response times, and negotiation strategy, versus the output measures, which include the number of rounds, total time, and ratio of the extra flight distances over extra baseline flight distances. The analysis results are relevant only to the specific CD&A algorithms that are used, but the same approach can be taken to analyze other CD&A algorithms. A statistical analysis is conducted across all combinations using a nonparametric Kruskal–Wallis H test to analyze the significance of the relationship between our input variables and output factors. The results of this statistical analysis help identify the possible factors that could affect negotiation.

A. Number of Rounds

The total number of negotiation rounds is recorded for each simulation run to determine how different conditions would impact the number of negotiation cycles. Statistical tests are conducted to evaluate the impact of the input variables on the number of rounds, and the significant results are reported below.

The number of rounds is averaged across different crossing angles and aircraft types. Figure 7 shows the results for the crossing angles. The number of rounds is the fewest for head-on scenarios, followed by regular crossing angle scenarios and shallow crossing scenarios in that sequence. The results suggest that the convergence of a conflict impacts the time pressure to complete the negotiations. Figure 8 shows the results for the aircraft type. The number of rounds is similar for the Global Hawk pairs and the Global Hawk–Sunglider combination, with the Sunglider pairs requiring nearly twice as many rounds. These results suggest that aircraft speed may also impact the number of rounds, possibly because slower aircraft can also impact the convergence.

Figure 9 shows the number of negotiation rounds for each response time pair. The results show a large number of rounds when both vehicle operators responded within 1 s of receiving the other operator's proposal. The results for the 1 s–1 s response time are

**Fig. 7** Number of rounds vs crossing angle.**Fig. 8** Number of rounds vs aircraft type.

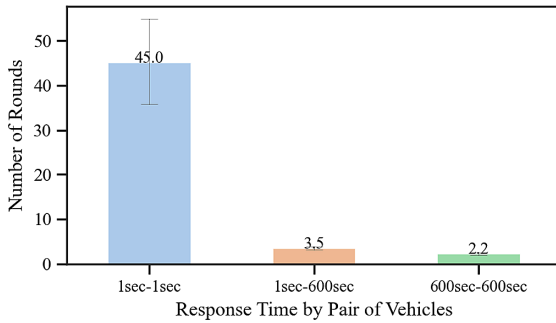


Fig. 9 Number of rounds vs response time.

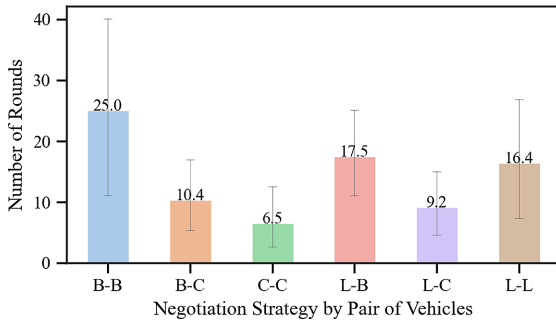


Fig. 10 Number of rounds vs negotiation strategy.

expected since shorter response times mean that operators can complete each round more quickly and have more overall time to negotiate with one another. As shown in Fig. 9, there is a significant reduction in the average number of rounds if any of the vehicles have a response time of 600 s instead of 1 s. Interestingly, only a few rounds are completed, even when one operator responds quickly while the other responds slowly. This suggests that multiple rounds of negotiations are only feasible when both vehicles can respond in rapid succession.

Figure 10 shows the number of negotiation rounds for different negotiation strategies by pairs of vehicles. The overall results show consistent patterns, where the number of rounds was the smallest when both vehicle operators used conceder strategies and the greatest when both used Boulware strategies. When the strategies are mixed, the number of rounds seems to vary with a general pattern: the conceder strategy results in fewer rounds, the Boulware strategy results in more, and the linear strategy is somewhere in between. These results suggest that when one or more vehicle operators “concede” and resolve more of the conflict earlier in the negotiation time frame, the negotiation tends to complete with fewer rounds, while the Boulware strategy has the opposite effect of extending the negotiation rounds by offering to resolve the conflicts later in the negotiation time frame. A further study of the interactions between the Boulware and conceder strategy pairs is explored in Sec. VII.D.

B. Total Time

The total negotiation time is calculated and examined across different conditions. The analysis suggested that all input variables impacted the length of the total time. Figure 11 shows the median total time for the different crossing angle types. Although the total time for the head-on angles is somewhat expected to be the shortest due to the fastest convergence, the results showed that the total time was the shortest for the regular crossing angles, followed by the head-on angles and then the shallow angles. Figure 12 shows the median total time for the different aircraft types. Different aircraft types significantly affect the total time, as the Global Hawk, which is an aircraft more maneuverable than the Sun glider, has nearly half the total time as the Sun glider pairs, which have the longest total time (see Fig. 12).

Figure 13 shows the total negotiation time for each response time pair. The total time is the shortest when both operators have a 1 s

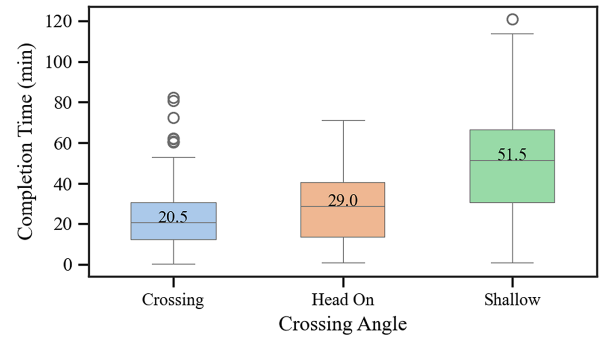


Fig. 11 Total time vs crossing angle.

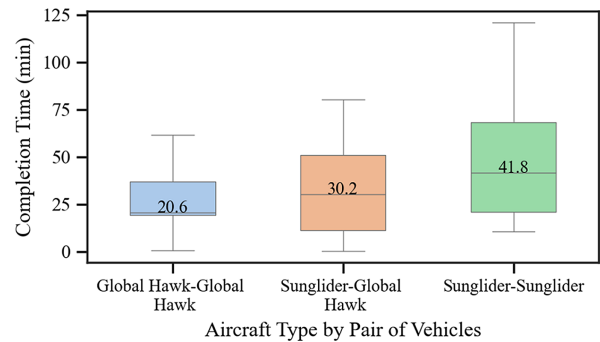


Fig. 12 Total time vs aircraft type.

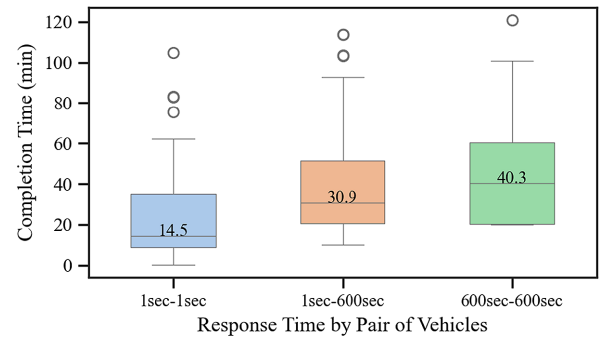


Fig. 13 Total time vs response time.

response time, and the median total time is the longest when both operators have a 600 s response time. This suggests that operators can reach a quicker mutual agreement if the response time is shorter.

The significance of the impact of negotiation strategy on total time can be observed in Fig. 14. The results align with our expectations of how the negotiation strategy would affect the negotiation. The Boulware-Boulware pair has the largest median value for total time, which is reasonable since the operator with the Boulware strategy is more hesitant to maneuver until the negotiation time

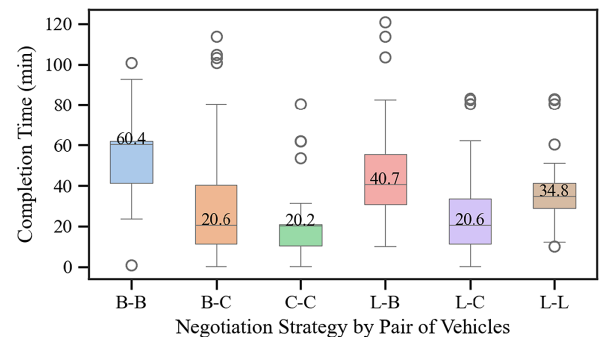


Fig. 14 Total time vs negotiation strategy.

window is low. Similarly, the conceder–conceder pair has the smallest median value for total time, likely due to operators maneuvering more aggressively early in the negotiation time window. Interestingly, any conditions in which one of the two operators employs the conceder strategy also result in a shorter total time, suggesting that early concession by any of the operators may result in a quick resolution to the conflict.

C. Ratio of the Extra Flight Distance Sum over Extra Baseline Flight Distance by A or B

In an ideal negotiation, two cooperative operators contribute a fair share of maneuver responsibility to resolve conflict. In such an outcome, the negotiated flight paths should be more efficient, resulting in less total extra distance flown than those without such negotiation. A direct measurement to assess the efficiency is to examine the sum of the extra flight distances flown by both vehicles in the negotiated solution, divided by the extra distance flown by each vehicle's corresponding baseline trajectory without the negotiation. This measurement returns a ratio score that describes both operators' added cooperative maneuver responsibility compared to what it would be if either operator took full maneuver responsibility. The data shown in Fig. 15 suggests that the total extra flight distance incurred by negotiation is generally far less than the extra distances incurred without negotiation for either operator A or B when one of the operators takes full responsibility. This benefits both operators since there is a high likelihood that the extra distance sum required will be less than that with either participant's full responsibility. It can be claimed that on an individual basis, each operator can reduce their total extra flight distance by an average of approximately 25% since the median ratio score for operators A and B resides around the 75% average. A 75% ratio sum means that the extra flight distance they flew together is about 75% of the extra flight distance compared to if any one of the vehicles flew the full amount. This is what happens on the median level, but we can observe from the overall bar that the extra flight distance can be greater than the full maneuver by either of the vehicles. The differing ratios between operators A and B are due to both operators having different flight plan geometry. As a result, the denominator values will vary between different crossing angles. Therefore, the implications of the results demonstrate that operators stand to benefit themselves and each other by participating in cooperative negotiation.

A potential loss in sum ratio efficiency is noted when comparing the sum ratio with different response times. From observations in Fig. 16, it can be seen that the 1 s–1 s response time pair (denoted by I–I for “instant–instant”) has the lowest median range, while the 600 s–600 s response time pair (denoted by L–L for “long–long”) has the highest median range. There is an observable increase in median scores as the response time pairs increase, and the ratio scores have a higher likelihood of exceeding the 100% threshold when the response times are longer. The general observation supports that faster response times result in a more efficient sum ratio, translating to less total distance flown for both vehicles.

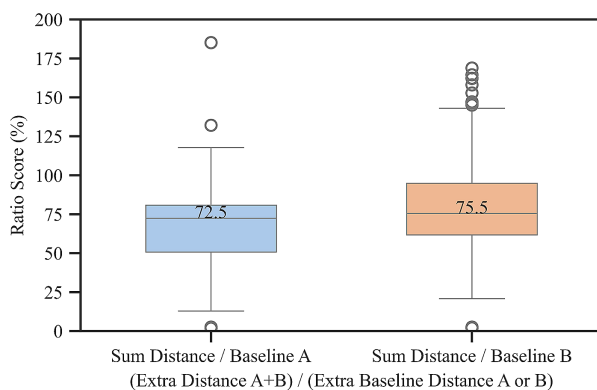


Fig. 15 Ratio of extra distance sum over baseline A or B.

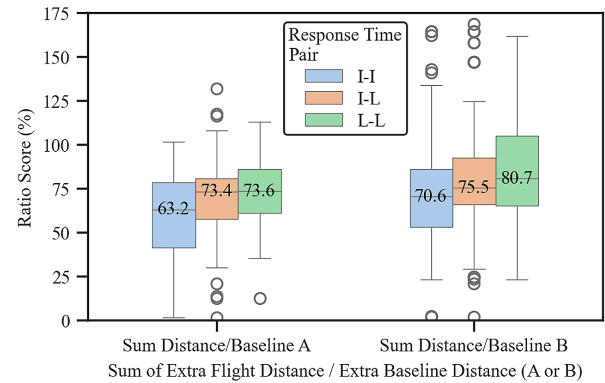


Fig. 16 Sum ratio vs response time.

D. Analyzing the Interactions for Boulware Versus Conceder Negotiation Strategies

One of the key questions of negotiation strategies for conflict resolution is the impact of the individual vehicles' strategies on the burden of the reroutes taken by the individual vehicles and the overall system. A question that arises from the sensitivity analysis is the potential interactions between the Boulware (B) and conceder (C) negotiation strategy pairs, whether giving in a large maneuver responsibility early on versus later in negotiation has a significant impact on the negotiation outcomes, and whether there are any significant advantages or disadvantages to any involved operators as a result of choosing Boulware or conceder strategies. This section describes the results of how the Boulware and conceder negotiation strategies affected the time utilization, the individual extra flight distance incurred, and the sum ratios for both operators.

The sum ratio overall trend in Fig. 17 shows less total cost when both parties do not concede than when both concede, with mismatching strategies (i.e., B–C pairs) showing similar or higher costs than the C–C condition, although the differences are relatively minor. Surprisingly, the mixed B–C and C–B strategies have equal or higher costs than the C–C strategy when both parties concede. The total costs, or the amount of partial maneuvers the vehicles have to move, are generally lower than the baseline cost of moving just one vehicle (approximately 76%) for all cases except for the C–B strategy for the B vehicle.

There was one exception to the patterns shown in Fig. 17, which is shown in Fig. 18. The exception is when the total sum ratio of the mismatched vehicle pairs (i.e., Sun glider–Global Hawk) that were rerouting collaboratively is compared to a baseline condition of moving just the fast vehicle (i.e., Global Hawk) during shallow angle conflicts. This result is likely due to the fact that in these vehicle pairs at shallow angles, any maneuvering by the Sun glider to resolve the conflict was far less efficient than the Global Hawk

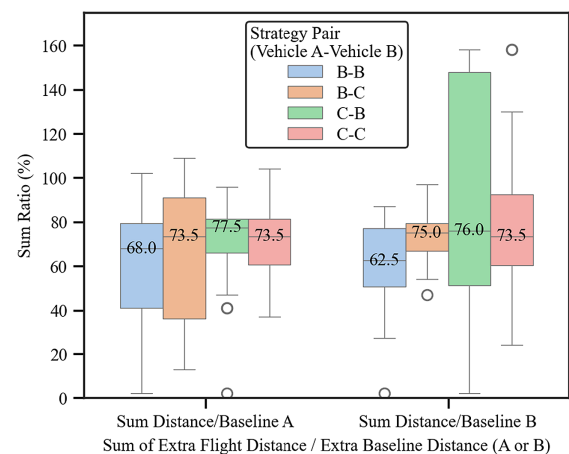


Fig. 17 Sum ratio by strategy pair.

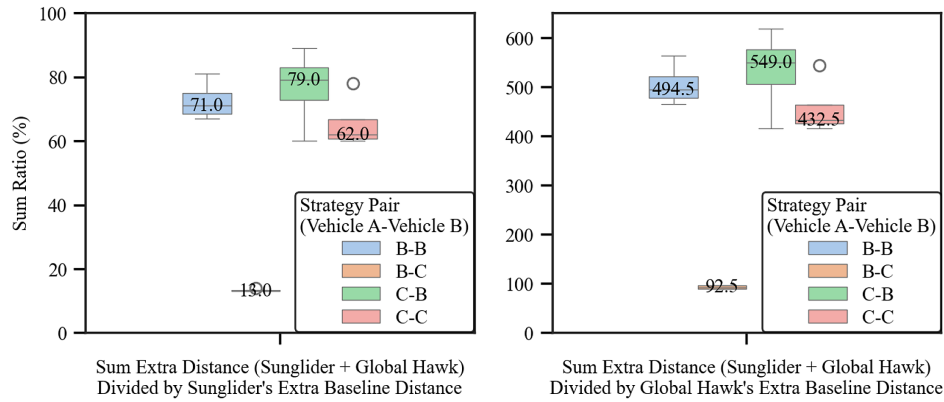


Fig. 18 Sum ratio of Sunglider or Global Hawk in shallow crossing angles by strategy pairs.

resolving the entire conflict alone, since the distance required to resolve the conflict is far greater than the calculated baseline distance. Therefore, both left and right figures in Fig. 17 had a much smaller sum ratio in the B–C condition (i.e., Global Hawk conceded and resolved most of the conflict) than in other conditions. In addition, the comparison of left versus right figures shows the same patterns of sum ratio, but the actual values were less than 100% on the left figure and far greater than 100% on the right figure. These results suggest that all the negotiated solutions resulted in less total flight distance than the solution by Sunglider alone, while they resulted in far greater flight distance than the solution by Global Hawk alone. These results suggest that moving the more maneuverable vehicle along under certain conditions is better than negotiating between the paired vehicles. Further research is needed to investigate other conditions when negotiation is less efficient.

Even though the B–B pairs result in a lower sum ratio score than the C–C pairs and mixed pairs, it does take longer to come to a negotiated solution using the B–B strategy than the C–C strategy (approximately 60% of total available negotiation completion time instead of about 20%), as shown in Fig. 19. Similarly, Fig. 20 shows that the number of rounds is much larger for the B–B strategy than for the C–C strategy. In a physical sense, this means that both operators spent more time negotiating when they both used the Boulware strategy and less time when they both used the conceder strategy.

As observed in Fig. 21, the B–B pairs result in less total cost than the C–C pairs; however, if the strategies are mismatched, then the conceder vehicle is burdened with a disproportionate amount of the cost or extra flight distance. The mismatched pairs mirror themselves in Fig. 21, where if vehicle A conceded, vehicle B with the Boulware strategy takes substantially less cost, and the same logic follows when the strategy pairs flip, where the vehicle that concedes takes on the substantially higher cost. It benefits the Boulware vehicle, as it flies less distance to resolve conflict, even though

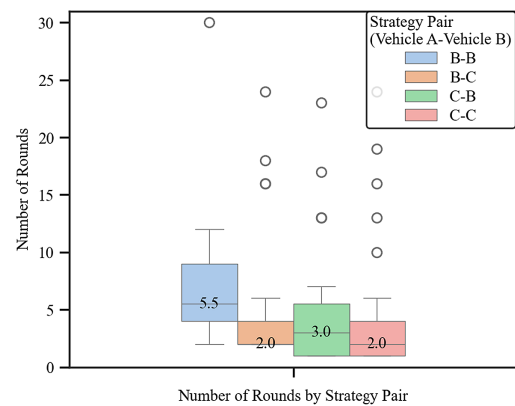


Fig. 20 Number of rounds by strategy pair.

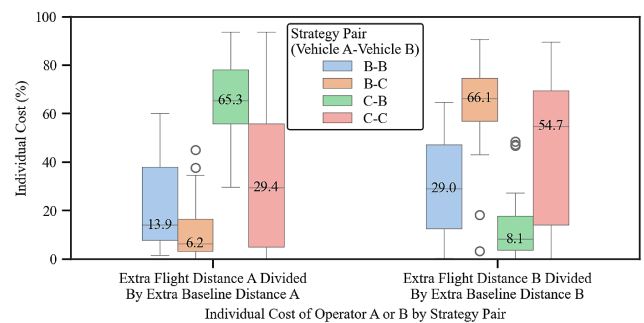


Fig. 21 Individual costs by strategy pair.

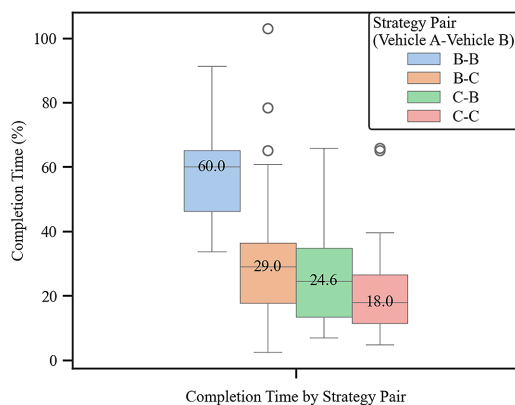


Fig. 19 Completion time by strategy pair.

the total cost across the two vehicles is comparable to other conditions, as shown in Fig. 22.

The analysis highlights a noticeable trend across various input parameters, including different angle types, response times, and aircraft pairings, significantly impacting key output measures such as the completion time, sum ratios, and the extra flight distance incurred by both operators. The findings suggest that the best condition for achieving favorable results occurs when both operators opt not to concede throughout the negotiation. Final resolutions in which only one or both operators concede early in the negotiation tend toward producing the least favorable outcomes. Negotiations for cases where there is a mismatch in negotiation strategies between two operators result in ambiguous outcomes contextual to the conflict scenario. There is also a recognition of potential game theory-based mechanics based upon initial results and possible questions about how an operator can effectively game the model to their benefit. Future testing and analysis will look into the theoretical game theory potential that can arise from different negotiation strategies.

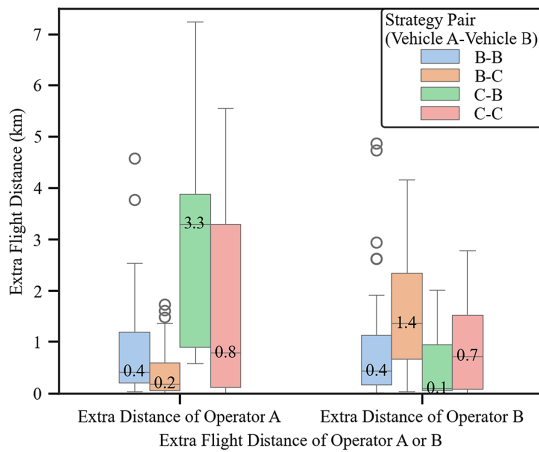


Fig. 22 Individual extra flight distance by strategy pair.

VIII. Key Observations

The study results provide multiple insights into how the negotiation strategies and operator response times affect the efficiency of the final negotiated flight trajectories and whether the negotiation paradigm benefits each operator. Furthermore, additional input variables on the crossing angles and the vehicle types are examined for the potential sensitivity of the results based on these parameters. Overall, the following are the key observations that were gathered from the study:

1) All experiments end with a successful, mutually agreed-upon conflict resolution, demonstrating the tool's effectiveness in solving conflicts with various input variables. The sensitivity analysis also suggests that participants will likely resolve conflict with a collaborative contribution ratio score sum of around 75% of the non-negotiation conflict resolution. Initial data suggests that a negotiated outcome between two parties results in a 25% reduction in total extra flight distance compared to if one participant took on the full maneuver. Most output variables were affected by the input variables.

2) Shorter response times are beneficial to all negotiating participants. It enables more rounds to process within the negotiation, directly correlating to a higher likelihood of a decreased contribution sum needed to resolve conflict. In this context, the larger number of negotiation rounds is considered to be beneficial, but there may also be a potential for creating an excessive workload due to faster cycles of negotiation. Overall, shorter response times benefit cooperative participants since they directly correlate to shorter flight distances and/or time required to resolve conflicts. Longer response times are problematic, leading to less efficient negotiations, with increased final contribution sums, longer completion times, and fewer numbers of rounds.

3) Aircraft types and crossing angles impact the output variables with similar trends. Factors that could potentially lengthen the time to conflict or decrease convergence, such as slow vehicle pairs (e.g., Sun glider–Sun glider pair) or shallow crossing angles, generally increased the total time and number of rounds to reach the completion of the negotiation. Faster vehicle pairs (e.g., Global Hawk–Global Hawk pair) and nonshallow crossing angles generally result in shorter completion time and fewer negotiation rounds, with the caveat that the regular crossing angles produce shorter total time than head-on crossings. Further research is needed to fully understand the complex interactions of the vehicles and crossing angles.

4) Total time, on a median level, takes roughly 20–60 min to resolve conflict through our negotiation model across various configurations, including crossing angles, aircraft types, response times, and negotiation strategies. A key point to note is that the completion time can be dependent on the processing speed of the hardware used, as well as the latency of the web-based tool, both of which require a more comprehensive analysis through a controlled study. While each negotiation ended successfully, with most having ample time left, further research is needed to determine the most impactful conditions that can cause the negotiations to run out of time.

5) Conceder strategies are more likely to reduce the total time spent on negotiation and the number of negotiation rounds needed to come to a resolution. This is reasonable since an earlier concession that offers a greater portion of the resolutions should allow quicker resolution with fewer negotiation rounds. Boulware strategies have the longest total times and the largest number of negotiation rounds before reaching a resolution and tend to resolve conflict with the least amount of total extra distance.

6) Scenarios with Boulware and conceder negotiation strategy pairs suggest that Boulware–Boulware pairs might have some advantages in terms of minimizing the total extra flight distances. It also suggests that when vehicles have large speed differentials, there may be certain traffic scenarios in which it is less costly to move the fast vehicle rather than moving both. There is an equity issue if the pair of vehicles employs different strategies, in which the conceder vehicle has the potential to absorb most of the cost unless the other vehicle also decides to use the conceder strategy. Other mechanisms may need to be implemented to minimize the equity issue (incentivize conceder strategies in other ways, only allow matching strategies, etc.), which can be explored in future studies.

IX. Conclusions

This paper introduces the first interactive web-based program to facilitate negotiation, detect conflict, and generate alternative flight trajectory plans for operators in real-time. For facilitated negotiation to be successful, a platform that can flexibly handle different configurations is needed. We develop a web application that can take an operator's desire to maneuver as a measurement of a private cost and streamline the complexities of resolving conflict with our internal algorithms. We are able to stress-test the tool through an automation system that allows us to set up several controlled experiments with different input parameters. We observe the effectiveness of our program by measuring key metrics such as the number of rounds, the total time, and the ratio of extra flight distances over extra baseline flight distances. Not only is our tool able to successfully resolve conflict for every experiment, but we also observe the statistical significance of the impact of our input variables on our output measures. This gives us a better understanding of how certain relationships between the input variables and output measures can affect the overall efficiency of the outcomes. The initial results of the sensitivity analysis not only demonstrate the effectiveness of our negotiation model in a real-time setting but also show promising indications that operators can benefit from cooperative negotiation resolutions versus the conventional method of taking on a full maneuver to avoid conflict.

Although the initial outcomes offer insight into a means of collaborative negotiation in ETM, future improvements can still be implemented to enhance our negotiation model. We plan to improve several aspects by improving the user interface, adding winds into the trajectory models, testing more vehicles and varying speeds, testing more conditions to better understand the impact of negotiation strategies, and expanding the model to handle multiparty negotiations. We recognize that equity and trust are critical issues for ETM, and our approach provides a method for transparency in resolutions that may improve trust and allow the negotiators to identify inequitable solutions. However, further exploration can be done to identify the conditions in which inequity occurs. Once scenarios in which inequity occurs are identified, additional mitigation approaches may be explored to prevent situations that result in inequitable solutions or develop a system to distribute equity to different stakeholders over time. The cumulation of suggested improvements allows the negotiation models to simulate more complex environments and better understand how to effectively resolve more conflict scenarios in the ETM space.

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